

J-Mathematics

A Rigorous Formal Framework

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An Axiomatic Extension of Classical Mathematics

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Abstract

J-Mathematics is an axiomatic extension of classical mathematics introducing a dual-component numerical universe. Every J-number possesses both a finite component and an infinite coordinate, the latter encoding informational history.

This framework introduces two distinct notions of equality: Full Entity Equality, preserving complete mathematical identity, and Destinal Equality, identifying objects that terminate at the same value regardless of their histories.

The Universal Golden Formula provides a canonical method for evaluating admissible functions at infinitary arguments using only two finite evaluations. The theory preserves mathematical history through representation sets and information fibres.

This monograph establishes the foundational axioms, algebraic structure, analytical principles, and representation theory of J-Mathematics, presenting it as a coherent extension of ordinary real arithmetic.

Chapter 1: Foundations

1.1 Primitive Objects and Syntax

1.1.1 Primitive Symbols

The following symbols are taken as primitive undefined terms of the theory.

1. $\mathbb{R}$, the set of real numbers, equipped with its standard field operations $+\mathbb{R}$ and $\cdot_{\mathbb{R}}$.
2. $0 \in \mathbb{R}$ and $1 \in \mathbb{R}$, denoting the additive and multiplicative identities of $\mathbb{R}$.
3. J , a distinguished symbol called the Infinite Unit.
4. $\mathcal{J}$, the universe of discourse for J-numbers.

1.1.2 J-Numbers and Coordinate Projections

The universe $\mathcal{J}$ is defined as the set of all formal expressions:

$$\mathcal{J} := \{a + bJ : a, b \in \mathbb{R}\}.$$

For every $x = a + bJ \in \mathcal{J}$, define the coordinate projection mappings:

$$\text{Fin} : \mathcal{J} \rightarrow \mathbb{R}, \quad \text{Fin}(x) := a,$$

$$\text{Inf} : \mathcal{J} \rightarrow \mathbb{R}, \quad \text{Inf}(x) := b.$$

The scalar a is the finite component; b is the infinite coordinate.

Axiom 1.1 (Coordinate Uniqueness). For all $x = a + bJ \in \mathcal{J}$ and $y = c + dJ \in \mathcal{J}$:

$$x = y \iff (a = c) \wedge (b = d).$$

1.1.3 Absolute Objects

The Absolute Origin is defined as:

$$\mathbf{0} := 0 + 0J.$$

The Absolute Infinity is defined as:

$$\mathbf{J} := 0 + 1J.$$

Axiom 1.2 (Infinite Unit Relation). The Infinite Unit satisfies the primitive reciprocal relation with zero. This is not interpreted as ordinary real division but as a defining axiom:

$$J = 0^{-1}, \quad J^{-1} = 0.$$

1.2 The Dual Hierarchy of Equality

Definition 1.3 (Full Entity Equality). For $x, y \in \mathcal{J}$, Full Entity Equality, denoted by $=$, holds iff both coordinates are identical:

$$x = y \iff (\text{Fin}(x) = \text{Fin}(y)) \wedge (\text{Inf}(x) = \text{Inf}(y)).$$

This asserts complete mathematical identity, including history and construction.

Definition 1.4 (Destinational Equality). For $x, y \in \mathcal{J}$, Destinational Equality, denoted by $\equiv_D$, holds iff x and y terminate at the same mathematical value (i.e., they have the same destination), regardless of their histories or J-coordinates.

Equivalently, $x \equiv_D y$ if and only if $\text{Fin}(x) = \text{Fin}(y)$, since the finite component is the numerical value itself.

Remark: Two J-numbers with different infinite coordinates but the same finite component are destinationally equal because they represent the same numerical value. For example, the Leibniz series for π gives:

$$\pi \equiv_D 4 - \frac{4}{3}J.$$

Both represent the value π , but they have different histories and J-coordinates.

Proposition 1.5 (Implication Hierarchy). For all $x, y \in \mathcal{J}$:

$$x = y \implies x \equiv_D y.$$

Proof: If $x = y$, then by Full Entity Equality, $\text{Fin}(x) = \text{Fin}(y)$. Hence, by Destinational Equality, $x \equiv_D y$. $\square$

Proposition 1.6 (Strictness of Hierarchy). The converse is false. That is:

$$\exists x, y \in \mathcal{J} : (x \equiv_D y) \wedge (x \neq y).$$

Proof: Construct $x = 5J = 0 + 5J$ and $y = 2J = 0 + 2J$. Clearly, $\text{Fin}(x) = 0 = \text{Fin}(y)$, so both represent the same value (0). Thus $x \equiv_D y$. However, $\text{Inf}(x) = 5 \neq 2 = \text{Inf}(y)$. By Coordinate Uniqueness, $x \neq y$. $\square$

Theorem 1.7 (Equivalence Relation). $\equiv_D$ is an equivalence relation on $\mathcal{J}$.

Proof:

1. *Reflexive:* For all x , $\text{Fin}(x) = \text{Fin}(x)$, so $x \equiv_D x$.
2. *Symmetric:* If $x \equiv_D y$, then $\text{Fin}(x) = \text{Fin}(y)$. Symmetry of equality in $\mathbb{R}$ gives $\text{Fin}(y) = \text{Fin}(x)$, so $y \equiv_D x$.
3. *Transitive:* If $x \equiv_D y$ and $y \equiv_D z$, then $\text{Fin}(x) = \text{Fin}(y)$ and $\text{Fin}(y) = \text{Fin}(z)$. Thus $\text{Fin}(x) = \text{Fin}(z)$, so $x \equiv_D z$. $\square$

1.3 Canonical Embedding of the Reals

Definition 1.8 (Canonical Embedding). Define the mapping $\iota : \mathbb{R} \rightarrow \mathcal{J}$ by:

$$\iota(r) := r + 0J.$$

The image $\mathbb{R}_{\mathcal{J}} := \iota(\mathbb{R}) = \{r + 0J : r \in \mathbb{R}\}$ is called the embedded copy of the real numbers in $\mathcal{J}$.

Proposition 1.9 (Injectivity of Embedding). ι is an injective mapping.

Proof: Suppose $\iota(r) = \iota(s)$. Then $r + 0J = s + 0J$. By Axiom 1.1 (Coordinate Uniqueness), comparing the finite coordinates gives $r = s$. Hence, ι is injective. $\square$

Chapter 2: Algebraic Operations

2.1 Addition and its Group Structure

Definition 2.1 (Addition). For $x = a + bJ \in \mathcal{J}$ and $y = c + dJ \in \mathcal{J}$, define the binary operation of addition $+$ on $\mathcal{J}$ by:

$$x + y := (a + c) + (b + d)J.$$

Theorem 2.2 (Abelian Group Structure of Addition). *The structure $(\mathcal{J}, +)$ is an abelian group.*

Proof:

1. *Closure:* For $a, c, b, d \in \mathbb{R}$, $a + c \in \mathbb{R}$ and $b + d \in \mathbb{R}$. Hence $(a + c) + (b + d)J \in \mathcal{J}$.
2. *Associativity:* Let $x = a + bJ$, $y = c + dJ$, $z = e + fJ$. Then:

$$(x + y) + z = ((a + c) + e) + ((b + d) + f)J = (a + (c + e)) + (b + (d + f))J = x + (y + z).$$

3. *Identity Element:* The Absolute Origin $\mathbf{0} = 0 + 0J$ satisfies $x + \mathbf{0} = (a + 0) + (b + 0)J = a + bJ = x$, and similarly $\mathbf{0} + x = x$.
4. *Inverse Element:* For $x = a + bJ$, define $-x := (-a) + (-b)J$. Then:

$$x + (-x) = (a - a) + (b - b)J = 0 + 0J = \mathbf{0}.$$

5. *Commutativity:*

$$x + y = (a + c) + (b + d)J = (c + a) + (d + b)J = y + x.$$

Thus, $(\mathcal{J}, +)$ satisfies all group axioms and is abelian. $\square$

2.2 The Zero Multiplication Axiom

Axiom 2.3 (Zero Extraction). Multiplication by the Absolute Origin extracts the infinite coordinate. For all $x = a + bJ \in \mathcal{J}$:

$$\mathbf{0} \cdot x = b, \quad x \cdot \mathbf{0} = b.$$

(Note: The results b are identified with their embedded images $b + 0J \in \mathcal{J}$ when necessary.)

2.3 Piecewise Definition of Multiplication

Definition 2.4 (Multiplication of J-Numbers). For $x = a + bJ \in \mathcal{J}$ and $y = c + dJ \in \mathcal{J}$, define the product $x \cdot y$ by the following piecewise rule:

$$x \cdot y := \begin{cases} b, & \text{if } y = \mathbf{0} \text{ (i.e., } c = 0 \text{ and } d = 0) \\ d, & \text{if } x = \mathbf{0} \text{ (i.e., } a = 0 \text{ and } b = 0) \\ ac + (ad + bc + bd)J, & \text{otherwise.} \end{cases}$$

(If both are zero, either case returns 0, so the definition is consistent.)

2.4 Basic Multiplicative Properties

Theorem 2.5 (Closure under Multiplication). *For all $x, y \in \mathcal{J}$, the product $x \cdot y$ belongs to $\mathcal{J}$.*

Proof: *If either is zero, the product returns a real number, identified with $r + 0J \in \mathcal{J}$. Otherwise, $ac \in \mathbb{R}$ and $ad + bc + bd \in \mathbb{R}$, so the general form belongs to $\mathcal{J}$. $\square$*

Theorem 2.6 (Commutativity of Multiplication). *For all $x, y \in \mathcal{J}$, $x \cdot y = y \cdot x$.*

Proof:

1. *If $x = \mathbf{0}$, then $x \cdot y = \text{Inf}(y)$ and $y \cdot x = \text{Inf}(y)$, so they are equal.*
2. *If $y = \mathbf{0}$, the argument is symmetric.*
3. *Otherwise, let $x = a + bJ$, $y = c + dJ$. Then:*

$$x \cdot y = ac + (ad + bc + bd)J,$$

$$y \cdot x = ca + (cb + da + db)J = ac + (bc + ad + bd)J.$$

Since addition and multiplication in $\mathbb{R}$ are commutative, the expressions are identical. $\square$

Theorem 2.7 (Multiplicative Identity). *The embedded real number $1 = 1 + 0J$ satisfies for all $x \in \mathcal{J}$:*

$$1 \cdot x = x \cdot 1 = x.$$

Proof: Since $1 \neq \mathbf{0}$, use the general case. Let $x = a + bJ$. Then:

$$1 \cdot x = (1 + 0J)(a + bJ) = 1 \cdot a + (1 \cdot b + 0 \cdot a + 0 \cdot b)J = a + bJ = x.$$

Commutativity gives $x \cdot 1 = x$. $\square$

2.5 Formal Failure of Associativity

Theorem 2.8 (Non-Associativity Counterexample). *There exist $x, y, z \in \mathcal{J}$ such that:*

$$(x \cdot y) \cdot z \neq x \cdot (y \cdot z).$$

Proof: Choose $x = J$, $y = J$, $z = \mathbf{0}$.

First, compute the left-hand side:

$$x \cdot y = J \cdot J = (0 + 1J)(0 + 1J).$$

Since neither is zero, apply the general rule:

$$J \cdot J = (0 \cdot 0) + (0 \cdot 1 + 1 \cdot 0 + 1 \cdot 1)J = 0 + (0 + 0 + 1)J = J.$$

Now multiply by $z = \mathbf{0}$:

$$(x \cdot y) \cdot z = J \cdot \mathbf{0} = \text{Inf}(J) = 1.$$

Now, compute the right-hand side:

$$y \cdot z = J \cdot \mathbf{0} = \text{Inf}(J) = 1.$$

Now multiply x by this result:

$$x \cdot (y \cdot z) = J \cdot 1.$$

Since $J \neq \mathbf{0}$ and $1 \neq \mathbf{0}$, apply the general rule:

$$J \cdot 1 = (0 + 1J)(1 + 0J) = 0 \cdot 1 + (0 \cdot 0 + 1 \cdot 1 + 1 \cdot 0)J = 0 + (0 + 1 + 0)J = J.$$

Since $1 \neq J$ (by Coordinate Uniqueness), we have:

$$(x \cdot y) \cdot z = 1 \neq J = x \cdot (y \cdot z).$$

Thus, associativity fails unconditionally. $\square$

Corollary 2.9 (Conditional Associativity). *Associativity holds if and only if no intermediate product invokes the zero-extraction rule. If all factors are non-zero, the general rule reduces to multiplication in the ring $\mathbb{R}[J]/(J^2 - J)$, which is associative. $\square$*

2.6 Formal Failure of Distributivity

Theorem 2.10 (Non-Distributivity Counterexample). *There exist $x, y, z \in \mathcal{J}$ such that:*

$$x \cdot (y + z) \neq x \cdot y + x \cdot z.$$

Proof: Choose $x = J$, $y = 1 + 0J$, $z = -1 + 0J$.

First, compute the sum:

$$y + z = (1 + 0J) + (-1 + 0J) = (1 - 1) + (0 + 0)J = \mathbf{0}.$$

Compute the left-hand side:

$$x \cdot (y + z) = J \cdot \mathbf{0} = \text{Inf}(J) = 1.$$

Now, compute the right-hand side. Since neither J nor 1 is zero:

$$J \cdot 1 = (0 + 1J)(1 + 0J) = 0 \cdot 1 + (0 \cdot 0 + 1 \cdot 1 + 1 \cdot 0)J = J.$$

Since $J \neq \mathbf{0}$ and $-1 \neq \mathbf{0}$:

$$J \cdot (-1) = (0 + 1J)(-1 + 0J) = 0 \cdot (-1) + (0 \cdot 0 + 1 \cdot (-1) + 1 \cdot 0)J = -J.$$

Thus:

$$x \cdot y + x \cdot z = J + (-J) = \mathbf{0}.$$

Since $1 \neq \mathbf{0}$:

$$x \cdot (y + z) = 1 \neq 0 = x \cdot y + x \cdot z.$$

Thus, distributivity fails unconditionally. $\square$

Corollary 2.11 (Conditional Distributivity). *Distributivity holds only if $\text{Inf}(x) = 0$ (i.e., x is an embedded real) or if $y + z \neq \mathbf{0}$. If both conditions fail (non-zero infinite coordinate and $y + z = \mathbf{0}$), the identity fails by the zero-extraction rule. $\square$*

Chapter 3: The Universal Golden Formula

3.1 J-Admissible Functions

Definition 3.1 (Admissibility). A function $F : \mathcal{D} \rightarrow \mathcal{J}$ is J-admissible if:

1. $\mathbb{R} \subset \mathcal{D} \subseteq \mathcal{J}$,
2. F is well-defined on its domain,
3. Every algebraic operation used in its definition is either addition or the piecewise multiplication from Chapter 2.

Proposition 3.2 (Power-Series Representation). *Every J-admissible function F admits a convergent power-series expansion:*

$$F(x) = \sum_{n=0}^{\infty} c_n x^n$$

on its domain, where $c_n \in \mathcal{J}$.

Proof: By Definition 3.1, F is built from constants, the variable x , and the operations of addition and multiplication. Any finite construction yields a polynomial. Any infinite construction is interpreted as the limit of its finite partial sums, which are polynomials; by the admissibility condition requiring validity in the region of evaluation, this limit converges to a power series. Hence the result follows. $\square$

3.2 Derivation of the UGF

Theorem 3.3 (Universal Golden Formula). *For every J-admissible function F :*

$$F(J) = F(0) + [F(1) - F(0)] \cdot J.$$

Proof: Let $F(x) = A + Bx$, where $A = a_1 + a_2J$ and $B = b_1 + b_2J$.

First, evaluate $F(0)$. By the Zero Extraction Axiom, $B \cdot 0 = b_2$. Thus:

$$F(0) = A + B \cdot 0 = (a_1 + a_2J) + b_2 = a_1 + (a_2 + b_2)J.$$

Next, evaluate $F(1)$. Since $1 \neq \mathbf{0}$, use the general multiplication rule:

$$B \cdot 1 = (b_1 + b_2J)(1 + 0J) = b_1 \cdot 1 + (b_1 \cdot 0 + b_2 \cdot 1 + b_2 \cdot 0)J = b_1 + b_2J.$$

Thus:

$$F(1) = A + B = (a_1 + b_1) + (a_2 + b_2)J.$$

Compute the difference:

$$F(1) - F(0) = [(a_1 + b_1) - a_1] + [(a_2 + b_2) - (a_2 + b_2)]J = b_1 + 0J = b_1.$$

Now evaluate $F(J)$. Since $J = 0 + 1J$ and $B \neq \mathbf{0}$:

$$B \cdot J = (b_1 + b_2J)(0 + 1J) = b_1 \cdot 0 + (b_1 \cdot 1 + b_2 \cdot 0 + b_2 \cdot 1)J = (b_1 + b_2)J.$$

Therefore:

$$F(J) = A + (b_1 + b_2)J = a_1 + (a_2 + b_1 + b_2)J.$$

Now compute the RHS of the UGF:

$$F(0) + [F(1) - F(0)]J = [a_1 + (a_2 + b_2)J] + b_1J = a_1 + (a_2 + b_2 + b_1)J.$$

Since $a_2 + b_1 + b_2 = a_2 + b_2 + b_1$, both sides are equal. For general admissible functions, the result extends by Proposition 3.1 (Power-Series Representation). $\square$

Theorem 3.4 (Generalized UGF). For all $a, b \in \mathbb{R}$ and J -admissible F :

$$F(a + bJ) = F(a) + [F(a + b) - F(a)] \cdot J.$$

Proof: Apply Theorem 3.2 to the auxiliary function $G(x) = F(a + bx)$. $\square$

3.3 Surjectivity of the UGF

Axiom 3.5 (Universal Surjectivity). For every $y \in \mathcal{J}$, there exists at least one J -admissible function F such that:

$$F(J) = y.$$

Definition 3.6 (Generator Family). For $y \in \mathcal{J}$, define the set of generators:

$$\mathcal{G}(y) := \{F : F \text{ is } J\text{-admissible and } F(J) = y\}.$$

Theorem 3.7 (Non-Empty Generators). For every $y \in \mathcal{J}$, $\mathcal{G}(y) \neq \emptyset$.

Proof: Immediate by Axiom 3.4. $\square$

Corollary 3.8 (Coordinate Generation). For every $y = a + bJ \in \mathcal{J}$, there exists a J -admissible F such that:

$$F(0) = a, \quad F(1) - F(0) = b.$$

Proof: By Axiom 3.4, choose F with $F(J) = y$. By Theorem 3.2, $F(J) = F(0) + [F(1) - F(0)]J$. By Coordinate Uniqueness, $\text{Fin}(F(J)) = F(0)$ must equal a , and $\text{Inf}(F(J)) = F(1) - F(0)$ must equal b . $\square$

3.4 Substitution of Destinationally Equal Representations

Theorem 3.9 (History-Preserving Substitution). Let F be J -admissible. If $x, y \in \mathcal{J}$ such that $x \equiv_D y$, then:

$$F(x) \equiv_D F(y).$$

Proof: Since $x \equiv_D y$, by Definition 1.4, x and y terminate at the same value. Let this common value be $a \in \mathbb{R}$. Then $x = a + bJ$ and $y = a + cJ$ for some $b, c \in \mathbb{R}$. Applying the Generalized UGF:

$$F(x) = F(a + bJ) = F(a) + [F(a + b) - F(a)]J,$$

$$F(y) = F(a + cJ) = F(a) + [F(a + c) - F(a)]J.$$

Clearly, $\text{Fin}(F(x)) = F(a) = \text{Fin}(F(y))$. Hence, by Definition 1.4, $F(x) \equiv_D F(y)$. $\square$

Theorem 3.10 (Non-Identity of Substitution). If $x \equiv_D y$ but $x \neq y$, then it is not necessarily true that $F(x) = F(y)$.

Proof: Follows directly from the proof of Theorem 3.8. The infinite coordinates are $F(a + b) - F(a)$ and $F(a + c) - F(a)$. If these are unequal, then $F(x) \neq F(y)$ by Axiom 1.1, even though they share the same destination. $\square$

Chapter 4: Sequences and Series

4.1 Infinite Evaluation of Sequences

Definition 4.1 (Admissible Sequence). A sequence is a J-admissible function $a : \mathbb{N} \rightarrow \mathcal{J}$ defined by $a(n) = a_n$.

Theorem 4.2 (Infinite Term). *For every admissible sequence (a_n) :*

$$a(J) = a_0 + (a_1 - a_0)J.$$

Proof: Apply Theorem 3.2 directly to the function a . We have $a(0) = a_0$ and $a(1) = a_1$. Hence:

$$a(J) = a(0) + [a(1) - a(0)]J = a_0 + (a_1 - a_0)J.$$

□

Definition 4.3 (Infinite Term). The value $a(J) \in \mathcal{J}$ is called the Infinite Term of the sequence.

4.2 Infinite Evaluation of Series

Definition 4.4 (Partial Sum Function). For a sequence (a_n) , define the partial sum function $S : \mathbb{N} \rightarrow \mathcal{J}$ by:

$$S(n) := \sum_{k=0}^n a_k.$$

Theorem 4.5 (Infinite Series Evaluation). *For every admissible sequence (a_n) with partial sum function S :*

$$S(J) = a_0 + a_1J.$$

Proof: By Definition 4.4, $S(0) = a_0$. Also, $S(1) = a_0 + a_1$. Applying Theorem 3.2 to S :

$$S(J) = S(0) + [S(1) - S(0)]J = a_0 + [(a_0 + a_1) - a_0]J = a_0 + a_1J.$$

□

Corollary 4.6 (Finite Determinism). *The J -evaluation of a series depends exclusively on the first two terms a_0 and a_1 . All subsequent terms contribute nothing to the J -evaluation.*

□

Chapter 5: Analytic Theory

5.1 Polynomial Functions

Definition 5.1 (J-Polynomial). A J-polynomial is a function $P : \mathbb{R} \rightarrow \mathcal{J}$ of the form:

$$P(x) = \sum_{k=0}^n c_k x^k,$$

where $c_k \in \mathcal{J}$ for all k .

Theorem 5.2 (Evaluation of Polynomials). *For every J-polynomial $P(x) = \sum_{k=0}^n c_k x^k$:*

$$P(J) = c_0 + \left(\sum_{k=1}^n c_k \right) J.$$

Proof: $P(0) = c_0$. $P(1) = \sum_{k=0}^n c_k$. Thus $P(1) - P(0) = \sum_{k=1}^n c_k$. Substituting into Theorem 3.2 yields the result. $\square$

5.2 Power Series

Definition 5.3 (Admissible Power Series). An admissible power series is:

$$F(x) = \sum_{n=0}^{\infty} c_n x^n,$$

valid in a region containing 0 and 1.

Theorem 5.4 (Evaluation of Power Series). *For every admissible power series:*

$$F(J) = c_0 + \left(\sum_{n=1}^{\infty} c_n \right) J.$$

Proof: Substituting $x = J$:

$$F(J) = \sum_{n=0}^{\infty} c_n J^n = c_0 J^0 + \sum_{n=1}^{\infty} c_n J^n = c_0 \cdot 1 + \sum_{n=1}^{\infty} c_n J = c_0 + \left(\sum_{n=1}^{\infty} c_n \right) J.$$

□

5.3 Binomial Expansion

Theorem 5.5 (Binomial Theorem in J-Space). *For all $a, b \in \mathbb{R}$ and $n \in \mathbb{N}$:*

$$(a + bJ)^n = a^n + \left[\sum_{k=1}^n \binom{n}{k} a^{n-k} b^k \right] J.$$

Proof: *By the classical Binomial Theorem:*

$$(a + bJ)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} (bJ)^k.$$

For $k = 0$, the term is a^n . For $k \geq 1$, $J^k = J$, so $(bJ)^k = b^k J$. Therefore:

$$(a + bJ)^n = a^n + \sum_{k=1}^n \binom{n}{k} a^{n-k} b^k J = a^n + \left[\sum_{k=1}^n \binom{n}{k} a^{n-k} b^k \right] J.$$

□

5.4 Taylor Expansion at J

Theorem 5.6 (Taylor at J). *Let F be analytic at 0. Then:*

$$F(J) = F(0) + \left(\sum_{n=1}^{\infty} \frac{F^{(n)}(0)}{n!} \right) J.$$

Proof: *Follows from Theorem 5.4 with $c_n = \frac{F^{(n)}(0)}{n!}$. □*

5.5 Taylor Expansion at J^{-1}

Theorem 5.7 (Taylor at J^{-1}). *Let F be analytic at x . Then:*

$$F(x + J^{-1}) = F(x) + \left[\sum_{n=1}^{\infty} \frac{F^{(n)}(x)}{n!} \right] J^{-1}.$$

Proof: By Taylor expansion:

$$F(x+h) = \sum_{n=0}^{\infty} \frac{F^{(n)}(x)}{n!} h^n.$$

Set $h = J^{-1}$. Since $n \geq 1$ implies $(J^{-1})^n = J^{-1}$, the result follows. $\square$

Chapter 6: Representation Theory

6.1 Representation Sets

Definition 6.1 (Representation Set). For $x \in \mathcal{J}$, define the Representation Set $R(x) \subseteq \mathcal{J}$ as:

$$R(x) := \{y \in \mathcal{J} : y \equiv_D x\}.$$

Theorem 6.2 (Non-Empty Representations). For every $x \in \mathcal{J}$, $R(x) \neq \emptyset$.

Proof: By Theorem 3.6, there exists a generator F with $F(J) = x$. The value $F(J)$ is trivially destinationally equal to itself, so $x \in R(x)$. $\square$

Theorem 6.3 (Cardinality of Representation Sets). For every $x = a + bJ \in \mathcal{J}$, the Representation Set $R(x)$ is uncountably infinite.

Proof: By Definition 6.1, $R(x) = \{a + cJ : c \in \mathbb{R}\}$. The mapping $\phi : \mathbb{R} \rightarrow R(x)$ given by $\phi(c) = a + cJ$ is a bijection. Since $\mathbb{R}$ is uncountable, $R(x)$ is uncountable. $\square$

6.2 History Sets

Definition 6.4 (History Set). Let $\mathcal{C}$ be the collection of all admissible mathematical constructions. Define the History Set $H(x)$ by:

$$H(x) := \{C \in \mathcal{C} : \text{the evaluation of } C \text{ terminates at } x\}.$$

Theorem 6.5 (Non-Empty Histories). For every $x \in \mathcal{J}$, $H(x) \neq \emptyset$.

Proof: By Axiom 3.4, there exists a J -admissible function F with $F(J) = x$. This function constitutes an admissible construction, hence belongs to $H(x)$. $\square$

Corollary 6.6 (Fibre Structure). Every destination x has an information fibre $H(x)$ consisting of all possible origins (histories) leading to x . $\square$

Chapter 7: Structural Algebra

7.1 Summary of Algebraic Properties

Theorem 7.1 (Structural Classification). *The algebra $(\mathcal{J}, +, \cdot)$ satisfies:*

1. **Addition:** *Abelian group.*
2. **Closure:** *Multiplication is closed.*
3. **Commutativity:** *Multiplication is commutative.*
4. **Identity:** *1 is the multiplicative identity.*
5. **Idempotence:** *$J^n = J$ for $n \geq 1$.*
6. **Associativity:** *Fails (explicit counterexample in Theorem 2.8).*
7. **Distributivity:** *Fails (explicit counterexample in Theorem 2.10).*

Proof: *Points (1)-(5) from earlier theorems. Points (6)-(7) from counterexamples. $\square$*

7.2 Invertibility and Zero Divisors

Theorem 7.2 (Non-Invertibility of Pure Infinite). *For any $\alpha \in \mathbb{R} \setminus \{0\}$, αJ is not invertible.*

Proof: *Suppose αJ is invertible. Then $\exists y = c + dJ$ such that $(\alpha J) \cdot y = 1$. Compute:*

$$(\alpha J) \cdot y = (0 + \alpha J)(c + dJ).$$

If $y \neq \mathbf{0}$, the general rule applies:

$$= 0 \cdot c + (0 \cdot d + \alpha \cdot c + \alpha \cdot d)J = \alpha(c + d)J.$$

For this to equal $1 = 1 + 0J$, we require $\alpha(c + d) = 1$ and the finite part to be 1, which is impossible because the finite part is 0. If $y = \mathbf{0}$, the product is α , not 1. Thus, no inverse exists. $\square$

Theorem 7.3 (Existence of Zero Divisors). *For every $\alpha \in \mathbb{R} \setminus \{0\}$, αJ is a zero divisor.*

Proof: *Construct the non-zero element $y = 1 - J$. Compute:*

$$(\alpha J)(1 - J) = \alpha(J - J^2) = \alpha(J - J) = \alpha \cdot \mathbf{0} = 0.$$

Since $y = 1 - J \neq \mathbf{0}$ (its infinite coordinate is -1), we have identified a non-zero annihilator. Hence αJ is a zero divisor. $\square$

Chapter 8: Examples and Applications

8.1 Constant Functions

Let $F(x) = c$. Then:

$$F(J) = c + (c - c)J = c.$$

8.2 Identity Function

Let $F(x) = x$. Then:

$$F(J) = 0 + (1 - 0)J = J.$$

8.3 Linear Functions

Let $F(x) = mx + c$. Then:

$$F(J) = c + mJ.$$

8.4 Exponential Function

Let $F(x) = e^x$. Then:

$$F(J) = 1 + (e - 1)J.$$

8.5 Natural Logarithm

Let $F(x) = \ln(1 + x)$. Then:

$$F(J) = 0 + \ln(2)J.$$

8.6 Trigonometric Functions

$$\sin(J) = \sin(1)J, \quad \cos(J) = 1 + (\cos(1) - 1)J.$$

8.7 Geometric Series

For $S(n) = \sum_{k=0}^n r^k$:

$$S(J) = 1 + rJ.$$

8.8 Fibonacci Sequence

For $F_0 = 0, F_1 = 1$:

$$F(J) = J.$$

Chapter 9: Foundational Principles

9.1 Primitive Axioms Summary

1. **Axiom 1.1:** Coordinate Uniqueness.
2. **Axiom 1.2:** Infinite Unit Relation ($J = 0^{-1}$).
3. **Axiom 2.3:** Zero Extraction ($\mathbf{0} \cdot x = \text{Inf}(x)$).
4. **Axiom 3.4:** Surjectivity of the UGF.

9.2 Substitution Principle

Theorem 9.1 (Substitution of Destinationally Equal Arguments). *Let F be J -admissible. If $x \equiv_D y$, then:*

$$F(x) \equiv_D F(y).$$

Proof: This is Theorem 3.8. $\square$

9.3 The Equality Hierarchy Theorem

Theorem 9.2 (Strict Hierarchy). *The relation of Full Entity Equality is strictly finer than Destinational Equality:*

$$(=) \subsetneq (\equiv_D).$$

Proof: By Proposition 1.5 and Proposition 1.6. $\square$

9.4 Grand Structure Theorem

Theorem 9.3 (Fundamental Theorem of J-Mathematics). *Let $\mathfrak{J} = (\mathcal{J}, +, \cdot, =, \equiv_D)$ be the complete structure. Then:*

1. $\mathbb{R}$ embeds canonically as a substructure.

2. *Every element has a unique coordinate decomposition.*
3. *Every admissible function satisfies the UGF.*
4. *Every destination has at least one generating history.*
5. *Substitution of destinationally equal objects preserves terminal destinations but not necessarily full identities.*

Proof: *Synthesized from Theorems 1.1 through 9.2. $\square$*